

# **Pesticide exposure assessment of Chinese potato farmers and their families using silicone wristbands (max. 100 characters)**

## **(Max. 10 authors)**

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## **ABSTRACT (max. 2000 characters, incl. spaces)**

### **Background**

Worldwide, China is the largest potato producer and consumer. However, intensive potato production is associated with intensive pesticide use, a potential public health threat. Therefore, the objective of this study is to assess pesticide exposure of Chinese potato farmers and their families.

### **Methods**

In total, 42 Chinese potato farmers were included: 14 organic, 14 conventional and 14 integrated farmers. Silicone wristbands (SWBs) were used to measure occupational (SWBocc) and residential (SWBres) pesticide exposure over two weeks. Information on demographics, agricultural practices and pesticide use were collected through questionnaires. Generalised Linear Models were used to examine the relationship between farmer characteristics (e.g. agricultural group, treated area, etc.) and (1) the number of pesticide residues (PR) detected on SWBocc and SWBres and (2) the cumulative PR concentration on SWBocc.

### **Results**

A total of 168 SWBs (2 weeks \* 42 SWBocc/SWBres) were analysed for the presence of PR. Per farmer, a median of 14 (IQR=8) and 3 (IQR=5) PR were found on the SWBocc and SWBres, respectively. All farmers were exposed to a lower number of PR at home than on the field ( $\beta=-1.1$ ,  $p < 0.01$ ) and integrated farmers were exposed to a lower number of PR than conventional farmers ( $\beta=-0.4$ ,  $p < 0.05$ ). On average, the occupational cumulative PR exposure ( $\log \sum PR$ ) was 3.1, 4.1 and 4.4 ng/WB for the organic, conventional and integrated farmers, respectively. A larger treated area and a higher self-reported pesticide use were associated with a higher cumulative PR concentration ( $\beta=1.9$ ,  $p < 0.01$ ;  $\beta=0.3$ ,  $p < 0.01$ , respectively).

### **Conclusion**

Potato farmers in China are exposed to a multitude of pesticides. Differences in pesticide exposure indicate that modifying certain agricultural practices may help reduce pesticide exposure among potato farmers. These findings could aid in educating farmers on how to minimize pesticide exposure during application.

### **Main messages (max. 200 characters per message):**

- Potato farmers in China use a multitude of pesticides; typically, 14 different PR were detected on individual occupational wristbands.
- Differences in pesticide exposure indicate that modifying certain agricultural practices may help reduce pesticide exposure among potato farmers.